ALMOST ALL LOOPS HAVE MULTIPLICATION GROUP S_n

Abstract. Cameron proved that for almost all quasigroups of order n the left translations generate the symmetric or the alternating group, and Häggkvist and Janssen excluded the alternating group; Cameron asked whether the same holds for loops, where the argument breaks because the translation by the identity is trivial. We show that it does, by counting: the probability that the rows of a uniformly random reduced Latin square of order n generate a group not containing A_n is at most $e^{-(\log 2-o(1))n^2}$, and the group is S_n with probability $1-O(c^n)$ for some $c<1$. The inputs are the van der Waerden and Brégman bounds on the number of Latin squares, the elementary bounds of Babai and Pyber on the orders of primitive groups, and the Häggkvist–Janssen bound on Latin squares all of whose rows are even permutations; the classification of finite simple groups is not used. This settles the question, raised by Cameron and recorded by Cavenagh, Greenhill and Wanless as a consequence of their conjecture on the second row, and completes an argument sketched by Eberhard in 2016.

1. Introduction

Let Q be a quasigroup of order n on the set $[n]=\{1,\dots,n\}$, with Cayley table a Latin square L, and for $x\in Q$ let $L_x\colon y\mapsto xy$ be the left translation, the x-th row of L read as a permutation of $[n]$. The *left multiplication group* $\mathrm{Mlt}_\lambda(Q)=\langle L_x : x\in Q\rangle$ is a transitive subgroup of S_n, and the full multiplication group $\mathrm{Mlt}(Q)$, generated by all left and right translations, contains it. Cameron [3] showed that for almost all quasigroups (that is, for all but a vanishing proportion of the Latin squares of order n) the left translations generate S_n or A_n; Häggkvist and Janssen [6] then showed that the proportion of Latin squares all of whose rows are even permutations is at most c^n for some $c<1$, so that A_n occurs for a proportion at most c^n of quasigroups. In particular $\mathrm{Mlt}(Q)$ is 2-transitive, which in Smith's character theory of quasigroups [15] means that almost all quasigroups have trivial character theory ("rank 2").

Cameron's argument is short. A uniformly random Latin square has a uniformly random first row, and by a theorem of Łuczak and Pyber [8] a uniformly random permutation lies in no transitive proper subgroup of S_n other than A_n with probability $1-o(1)$; since the rows generate a transitive group, that group contains A_n.

For *loops* — quasigroups with an identity element, whose Cayley tables, suitably labelled, are the *reduced* Latin squares, with first row and first column in natural order — the argument is not available: the first row is the identity permutation, and no other row is uniformly distributed. Cameron asked whether almost all loops nevertheless have multiplication group S_n (see [4], and [5, §6], where Cavenagh, Greenhill and Wanless record, as Lemma 6.2, that this follows from their conjecture on the cycle structure of the second row of a random Latin square). The statement is folklore in loop theory, where it is mentioned as an unproved belief. The purpose of this note is to settle it, by an argument that uses no information about the second row at all.

Theorem 1. *Let L be a uniformly random reduced Latin square of order n and $H(L)\le S_n$ the group generated by its rows. Then*

$$\mathbb{P}\big[H(L)\not\supseteq A_n\big]\le e^{-(\log 2-o(1))n^2},\qquad \text{and}\qquad \mathbb{P}\big[H(L)=S_n\big]\ge 1-3c^n$$

for some absolute $c<1$ and all large n. Consequently almost all loops have $\mathrm{Mlt}_\lambda(Q)=\mathrm{Mlt}(Q)=S_n$ and trivial character theory.

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The idea, which Eberhard sketched in comments on Cameron's post [4], is that the event is about all $n-1$ non-identity rows simultaneously, and such events are cheap to bound by counting: for a subgroup $K \le S_n$, at most $|K|^{n-1}$ of the $L(n)/n! = (n!)^{n-1}e^{-n^2+o(n^2)}$ Latin squares with first row the identity have all their rows in K, a proportion of at most $e^{n^2+o(n^2)}/[S_n : K]^{n-1}$, which is $e^{-\delta n^2+o(n^2)}$ as soon as $[S_n : K] \ge e^{(1+\delta)n}$. A transitive proper subgroup lies in a maximal one, and the maximal transitive subgroups of S_n are the wreath products $S_a \wr S_b$, $ab = n$, of which there are fewer than $n \cdot n!$, and the primitive maximal subgroups. The crude count disposes of $S_a \wr S_b$ for $b \ge 3$ (the index $n!/((a!)^b b!)$ is a multinomial coefficient over $b!$, at least $3^n/n^{O(1)}$) but not of $S_{n/2} \wr S_2$ (index $\binom{n}{n/2}/2 < e^n$) nor of A_n; for the primitive subgroups one needs a bound on their orders strong enough that their number does not matter. We handle every imprimitive group by a single block decomposition (Lemma 2), which is sharper than the crude count — for $b = 3$ the crude count gives $e^{-(\log 3-1)n^2+o(n^2)}$ and for $b = 2$ nothing, while the lemma gives $e^{-(\log 2-o(1))n^2}$ for every b — and in particular disposes of $S_{n/2} \wr S_2$; the primitive groups by the elementary order bounds of Babai and Pyber, after which their number is controlled by counting generators; and A_n by the theorem of Häggkvist and Janssen. No step uses the classification of finite simple groups.

2. Preliminaries

Latin squares. Let $\mathcal{L}_n$ be the set of Latin squares of order n on the symbol set $[n]$ and $L(n) = |\mathcal{L}_n|$. We use two classical bounds [17, Ch. 17]: the lower bound $L(n) \ge (n!)^{2n}/n^{n^2}$, a consequence of the van der Waerden permanent inequality applied row by row, and the upper bound $L(n) \le \prod_{k=1}^{n}(k!)^{n/k}$, a consequence of Brégman's. In the explicit forms we need,

$$\log L(n) \ge n^2 \log n - 2n^2, \qquad \log L(a) \le a^2 \log a - 2a^2 + a\big(\tfrac{1}{2}\log^2 a + 2\log a + 3\big) \tag{1}$$

for all $n, a \ge 1$. The first follows from $n! \ge (n/e)^n$. For the second, $\log k! \le k\log k - k + \log k + 1$ for $k \ge 1$ gives $(\log k!)/k \le \log k - 1 + (\log k + 1)/k$, and summing over $k \le a$ with $\log a! \le a\log a - a + \log a + 1$, $\sum_{k\le a}(\log k)/k \le \frac{1}{2}\log^2 a + 1$ and $\sum_{k\le a} 1/k \le \log a + 1$,

$$\sum_{k=1}^{a} \frac{\log k!}{k} \le a\log a - 2a + \tfrac{1}{2}\log^2 a + 2\log a + 3,$$

which multiplied by a is the claim.

Loops and reduced squares. For $L \in \mathcal{L}_n$ with rows $r_1, \dots, r_n \in S_n$ (so $r_i(c) = L(i,c)$), relabelling the symbols by r_1^{-1} gives the square with rows $\sigma_i = r_1^{-1} r_i$, whose first row is the identity; this map $\mathcal{L}_n \to \mathcal{L}_n^0 := \{L : r_1 = \mathrm{id}\}$ is $n!$-to-one, so a proportion of squares in $\mathcal{L}_n^0$ is the same proportion of squares in $\mathcal{L}_n$, and $|\mathcal{L}_n^0| = L(n)/n!$; and relabelling the symbols by any $\tau \in S_n$ replaces each r_i by τr_i and so leaves every $r_1^{-1} r_i = (\tau r_1)^{-1}(\tau r_i)$, hence the group they generate, unchanged. Permuting the rows of L^0 so that its first column is $1, 2, \dots, n$ gives a reduced square; each reduced square arises from exactly $(n-1)!$ squares L^0, so proportions agree again, and the two have the same *set* of rows, with the identity row still first (it is the unique row of L^0 with entry 1 in column 1). A uniform reduced square is the Cayley table of a uniform random loop on $[n]$ with identity 1, whose left translations are its rows. Writing

$$H(L) := \langle \sigma_2, \dots, \sigma_n \rangle = \langle r_1^{-1} r_i : 2 \le i \le n \rangle \le S_n,$$

it follows that for every subgroup $K \le S_n$, the proportion of $L \in \mathcal{L}_n$ with $H(L) = K$ equals the proportion of loops Q of order n with $\mathrm{Mlt}_\lambda(Q) = K$, and it suffices to prove the bounds of Theorem 1 for $H(L)$ as proportions of $\mathcal{L}_n$, or equivalently of $\mathcal{L}_n^0$, which is what we do. $H(L)$ is transitive: for $c, s \in [n]$ the symbol $r_1(s)$ occurs in column c in some row i, and then $\sigma_i(c) = s$.

Group theory. We use three facts about subgroups of S_n. A transitive subgroup not containing A_n is either imprimitive, i.e. contained in the stabiliser $\mathrm{Stab}(\Pi) \cong S_a \wr S_b$ of a partition Π of $[n]$ into b blocks of size a with $1 < a < n$, or primitive. A primitive subgroup not containing A_n has order at most $e^{4\sqrt{n}\log^2 n}$ for $n \geq n_0$: for uniprimitive groups (primitive but not doubly transitive) this is Babai's theorem [1], and a doubly transitive group not containing A_n has order at most $\exp(\exp(c\sqrt{\log n}))$ (Babai [2]) and at most $n^{c\log^2 n}$ (Pyber [12]), both smaller for large n; the proofs of all three bounds are elementary, in particular they do not use the classification of finite simple groups. Finally a group of order m is generated by at most $\log_2 m$ elements (adjoining to a proper subgroup an element outside it at least doubles the order).

3. Proof of Theorem 1

Lemma 2 (squares with a block structure)**.** *Let $\Pi = \{C_1, \dots, C_b\}$ be a partition of the columns $[n]$ and $\Pi' = \{S_1, \dots, S_b\}$ a partition of the symbols $[n]$, both into b blocks of size a, $ab = n$. The number of Latin squares $L \in \mathcal{L}_n$ every row of which maps each block of Π onto a block of Π' is at most $(b!)^n\, L(a)^{b^2}$. (Only $\Pi' = \Pi$ is used below; the proof is the same for two partitions.)*

Proof. Each row r_i of such a square induces a bijection $\pi_i \colon [b] \to [b]$ with $r_i(C_j) = S_{\pi_i(j)}$. Fix j, j' and let $I_{jj'} = \{i : \pi_i(j) = j'\}$. In a column $c \in C_j$ each of the a symbols of $S_{j'}$ occurs exactly once, and only in rows of $I_{jj'}$, each of which contributes exactly one symbol of $S_{j'}$ to column c; hence $|I_{jj'}| = a$, and the subarray with rows $I_{jj'}$ and columns C_j is an $a \times a$ Latin square on the symbol set $S_{j'}$. Thus L is determined by the $n \times b$ array $(\pi_i(j))$, whose rows are permutations of $[b]$, together with one Latin square of order a for each of the b^2 pairs (j, j'), on the row set $I_{jj'}$, column set C_j and symbol set $S_{j'}$ determined by the array (identifying each of these sets with $[a]$ in increasing order). There are at most $(b!)^n$ arrays (those that occur satisfy further constraints, which we do not need) and $L(a)$ choices for each small square. □

Proof of Theorem 1. Throughout, $\mathbb{P}$ denotes proportion in $\mathcal{L}_n^0$; by the discussion above, these proportions equal the corresponding proportions of reduced squares, hence of loops. Let L^0 be uniform on $\mathcal{L}_n^0$ and $H = H(L^0)$. Since H is transitive, if $H \not\geq A_n$ then H is imprimitive or primitive of order at most $B := e^{4\sqrt{n}\log^2 n}$ (for $n \geq n_0$, by the group-theoretic facts above).

Imprimitive. Suppose H preserves a partition Π of $[n]$ into b blocks of size a, $1 < a < n$. Then every row of L^0 maps each block of Π onto a block of Π, so by Lemma 2 with $\Pi' = \Pi$,

$$\mathbb{P}\big[H \leq \mathrm{Stab}(\Pi)\big] \leq \frac{(b!)^n\, L(a)^{b^2}}{L(n)/n!}.$$

There are $n!/((a!)^b b!) \leq n!$ partitions Π with blocks of size a and fewer than n divisors a of n. By (1), with $b = n/a$, $(b!)^n \leq b^{nb}$ and $\log n! \leq n\log n$,

$$\begin{aligned}\log \frac{n!\cdot n!\,(b!)^n\, L(a)^{b^2}}{L(n)} &\leq 2n\log n + \frac{n^2}{a}\log b + \Big(n^2\log a - 2n^2 + \frac{n^2}{a}\big(\tfrac12\log^2 a + 2\log a + 3\big)\Big)\\ &\qquad - n^2\log n + 2n^2\\ &= -n^2\log b + \frac{n^2}{a}\Big(\log b + \tfrac12\log^2 a + 2\log a + 3\Big) + 2n\log n.\end{aligned}$$

If $a \geq \sqrt{n}$, the middle term is at most $n^{3/2}(\frac12\log^2 n + 3\log n + 3) = o(n^2)$, so the right side is $-n^2\log b + o(n^2) \leq -n^2\log 2 + o(n^2)$. If $2 \leq a < \sqrt{n}$, then $b > \sqrt{n}$, $(n^2/a)\log b \leq \frac12 n^2 \log b$, and $(\frac12\log^2 a + 2\log a + 3)/a \leq \frac52$ (the function is decreasing in $a \geq 1$ and is less than 2.32 at $a = 2$), so the right side is at most $-\frac12 n^2\log b + \frac52 n^2 + 2n\log n \leq -\frac14 n^2\log n + 3n^2$, which is at most

$-n^2\log 2$ once $n \ge e^{15}$. Summing over Π and a,

$$\mathbb{P}[H \text{ imprimitive}] \le n\, e^{-(\log 2 - o(1))n^2},$$

where the $o(1)$ is $O(\log^2 n/n)$: the term $a = n/2$ is at most $e^{-n^2\log 2 + O(n\log^2 n)}$, every term with $b \ge 3$ and $a \ge \sqrt{n}$ is at most $e^{-n^2\log 3 + O(n^{3/2}\log^2 n)}$, and every term with $a < \sqrt{n}$ is at most $e^{-n^2\log 2}$ for $n \ge e^{15}$, so the sum is at most $n\, e^{-n^2\log 2 + O(n\log^2 n)}$.

Primitive. Let $n \ge n_0$. For a fixed subgroup $K \le S_n$, the squares $L^0 \in \mathcal{L}_n^0$ all of whose rows lie in K number at most $|K|^{n-1}$. A subgroup of order at most B is generated by at most $\log_2 B \le 6\sqrt{n}\log^2 n$ elements, so there are at most $(n!)^{6\sqrt{n}\log^2 n} \le e^{6n^{3/2}\log^3 n}$ of them. Summing over these,

$$\begin{aligned}\mathbb{P}\big[H \text{ primitive},\ H \not\supseteq A_n\big] &\le \frac{e^{6n^{3/2}\log^3 n}\, B^{\,n-1}}{L(n)/n!}\\ &\le \exp\big(-n^2\log n + 2n^2 + n\log n + 6n^{3/2}\log^3 n + 4n^{3/2}\log^2 n\big)\\ &= e^{-(1-o(1))n^2\log n}.\end{aligned}$$

Adding the two cases gives the first inequality of the theorem.

The sign. $H \le A_n$ if and only if every row σ_i of L^0 is even. Under the $n!$-to-one map $L \mapsto L^0$ this is the condition that every row r_i of $L \in \mathcal{L}_n$ has the parity of r_1, so the proportion of $L^0 \in \mathcal{L}_n^0$ with all rows even equals the proportion of $L \in \mathcal{L}_n$ with all rows of one parity. Relabelling the symbols by a transposition is a bijection of $\mathcal{L}_n$ reversing the parity of every row, so the squares with all rows odd are as many as those with all rows even, and the latter are at most $c^n|\mathcal{L}_n|$ by Häggkvist and Janssen [6]. Hence $\mathbb{P}[H \le A_n] \le 2c^n$. If $H \supseteq A_n$ and $H \not\le A_n$ then $H = S_n$; so $\mathbb{P}[H \ne S_n] \le e^{-(\log 2 - o(1))n^2} + 2c^n \le 3c^n$ for large n. Finally $\mathrm{Mlt}(Q) \supseteq \mathrm{Mlt}_\lambda(Q) = S_n$ is 2-transitive, which is the rank-2 condition of [15]. □

4. Remarks

Remark 3 (sharpness). For even n the exponent $\log 2$ is attained, by the subgroup $S_{n/2} \wr S_2$ (the bound is a union bound, so "attained" means that this one term already has the exponent of the total): the squares all of whose rows preserve a fixed partition into two blocks of size $n/2$ number exactly $\binom{n}{n/2}L(n/2)^4$ — in the proof of Lemma 2 with $b = 2$ the array is determined by the $n/2$ rows that swap the blocks, and conversely every array of this kind together with any four small squares gives a Latin square — and $\binom{n}{n/2}L(n/2)^4/L(n) = 2^{-n^2+O(n\log^2 n)}$ by the asymptotics $\log L(m) = m^2\log m - 2m^2 + O(m\log^2 m)$ that (1) encodes (this counts squares with arbitrary first row; the proportion among those with first row the identity is larger by $\frac{1}{2}\binom{n}{n/2} = 2^{O(n)}$, which does not affect the exponent); the union over the $\frac{1}{2}\binom{n}{n/2}$ such partitions changes this by $2^{O(n)}$. For odd n the bound of the theorem is not sharp: with p the least prime factor of n, the partition into p blocks gives, by the same count (with the array in which the rows of the k-th block of n/p rows are the k-th row of the cyclic Latin square of order p, so that every choice of the p^2 small squares gives a Latin square), at least $L(n/p)^{p^2} = e^{-(\log p + o(1))n^2}L(n)$ squares with H imprimitive, and the proof gives $\mathbb{P}[H \not\supseteq A_n] \le e^{-(\log p - o(1))n^2}$ whenever $p \le n^{1/4}/e^3$ (so that the terms with $a < \sqrt{n}$ are smaller); for prime n every transitive group is primitive and $\mathbb{P}[H \not\supseteq A_n] \le e^{-(1-o(1))n^2\log n}$. The threshold $n \ge e^{15}$ in the proof is an artefact of the case split and has not been optimised: the sum over the divisors a of n of $n!\cdot n!\,(b!)^n L(a)^{b^2}/L(n)$ (the first display of the imprimitive case times the crude count $n!$ of partitions, with (1) inserted for $L(a)$ and $L(n)$ and $n!$, $(b!)^n$ exact) is already less than 1 for every composite $n \ge 93$ (and is $e^{-0.88\,n^2\log 2}$ at $n = 1000$), and with the exact van der Waerden and Brégman expressions

in place of (1) and the exact number $n!/((a!)^b b!)$ of partitions in place of $n!$ it is less than 1 for every composite $n \geq 15$ ($e^{-0.37\,n^2 \log 2}$ at $n = 30$, $e^{-0.70\,n^2 \log 2}$ at $n = 100$); these are finite computations [14], not uniform inequalities.

Remark 4 (quasigroups). With $G = \langle r_1, \dots, r_n \rangle$ in place of H, the same three steps (Lemma 2 with $\Pi' = \Pi$, the count $|K|^n$, and [6]) reprove Cameron's theorem [3] without [8], with the same bounds: $e^{-(\log 2 - o(1))n^2}$ for $G \not\supseteq A_n$ and $3c^n$ for $G \neq S_n$. The converse direction is not available: the loop case does not follow from the quasigroup case, because the symbol relabelling that produces a reduced square replaces the row group $\langle r_i \rangle$ by $\langle r_1^{-1} r_i \rangle$.

Remark 5 (the primitive case). The primitive case needs only that every primitive $K \not\supseteq A_n$ has $\log |K| = o(n)$: then the generator count makes the number of such K at most $(n!)^{o(n)} = e^{o(n^2 \log n)}$, negligible against $L(n)/n!$. Praeger and Saxl's elementary bound $|K| < 4^n$ [10] is too weak for this (it gives $(n!)^{2n}$ subgroups), which is why a first version of this note combined it with Pyber's bound $2^{(\frac{1}{6}\log_2 24 + o(1))n^2}$ on the number of subgroups of S_n [11] (recently sharpened to $2^{n^2/16 + O(n^{3/2})}$ by Roney-Dougal and Tracey [13]); we have not checked whether that count can be obtained without the classification of finite simple groups. Eberhard pointed out that the elementary order bounds of Babai and Pyber used above make the primitive case self-contained and visibly free of it. The bound $\exp(O(n^{1/3} \log^{7/3} n))$ of Sun and Wilmes [16] for uniprimitive groups, also proved without the classification, would do as well; anything with $\log |K| = o(n)$ does.

Remark 6 (what is and is not used). Nothing is used about the distribution of any single row. The event $H \leq K$ is the intersection of the $n-1$ events $\sigma_i \in K$, and the identity row, which defeats the one-row argument, contributes nothing to it. The crude count $|K|^{n-1}$ against $(n!)^{n-1} e^{-n^2 + o(n^2)}$ already settles each individual K of index at least $e^{(1+\delta)n}$; the two maximal subgroups of smaller index, A_n and $S_{n/2} \wr S_2$ (with their conjugates), are where the Latin constraints inside K have to be used, through [6] and through Lemma 2 respectively. In particular the theorem says nothing about the conjecture of [5] on the cycle type of the second row, which concerns one row given the identity row, and which remains open; the consequence recorded in [5, Lemma 6.2] simply does not need it.

Remark 7 (Eberhard's sketch). In comments on [4] (October 2016), Eberhard observed that the number of Latin squares with all rows in K is at most $|K|^n = (n!)^n/[S_n : K]^n$ against $L(n) = (n!)^n/e^{n^2 + o(n^2)}$, concluded that only A_n and $S_{n/2} \wr S_2$ need separate treatment, and asked whether the method of [6] adapts to other subgroups; Cameron replied that this "should not be too hard" and that the second-row conjecture was the most interesting part; Eberhard then sketched for $S_{n/2} \wr S_2$ a row-by-row count (fill the first $k \approx n/10$ rows inside the group, bound the rest by the standard row-by-row estimate). The sketch is correct once one passes to maximal subgroups, which takes care of the imprimitive groups with $b \geq 3$; what remains to be supplied is the primitive case, for which an elementary order bound for primitive groups suffices, as Eberhard pointed out to us (the sketch's "only A_n and $S_{n/2} \wr S_2$" presupposes such a bound; see Remark 5), and the details for $S_{n/2} \wr S_2$. Lemma 2 treats all imprimitive groups uniformly and gives the sharp exponent. As far as we know the argument was not written out.

Remark 8 (small orders). For $n = 5$, of the 56 reduced squares 50 have $H = S_5$ and 6 have $H = C_5$ (the sorted Cayley table of $\mathbb{Z}_5$ for each of the six cyclic subgroups; these are also the all-even ones). For $n = 6$, of 9408 reduced squares 7776 have $H = S_6$, 72 have A_6 and 1560 have an imprimitive H (block sizes 2 and 3, orders 6 to 48). In samples of 3000 Latin squares at each order $7 \leq n \leq 16$ produced by the Jacobson–Matthews chain [7] (the chain on proper and improper squares, started from the cyclic square, run for $200t$ moves and then recorded

at those times $200t + kt$, $k \geq 1$, at which the state is proper, improper checkpoints being skipped with the grid kept, $t = 20n^2$; the stationary distribution restricted to proper states is uniform and this fixed-grid sampling is unbiased, but we have not verified mixing, so these are samples from a chain, not certified uniform samples), the number of squares with $H = A_n$ was $37, 25, 18, 4, 1, 4, 0, 1, 0, 0$ for $n = 7, \ldots, 16$, and every other H was S_n except for two squares at $n = 8$ with $H = S_2 \wr S_4$ (order 384) [14].

Remark 9 (labelled and unlabelled loops). The theorem counts labelled loops on $[n]$ with identity 1, i.e. reduced squares, and the error bound survives the passage to isomorphism classes. Let $p_n \leq 3c^n$ be the proportion of reduced squares with $H \neq S_n$ and ε_n the proportion of reduced squares whose loop has a nontrivial automorphism group. A loop automorphism is an autotopism of the square, the autotopy group is an isotopy invariant, and the map from $\mathcal{L}_n$ to reduced squares is uniform, so ε_n is at most the proportion of Latin squares with a nontrivial autoparatopy group, which is $n^{-3n^2/8+o(n^2)}$ by [9, Theorem 4]. An isomorphism class with trivial automorphism group contains exactly $(n-1)!$ reduced squares, and every class contains at least one, so the proportion of isomorphism classes with multiplication group $\neq S_n$ is at most $(p_n + (n-1)!\,\varepsilon_n)/(1-\varepsilon_n) = O(c^n)$.

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